ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Real and Reciprocal Spaces I

I. The Geometry and Art of Real Space for Crystal Structures

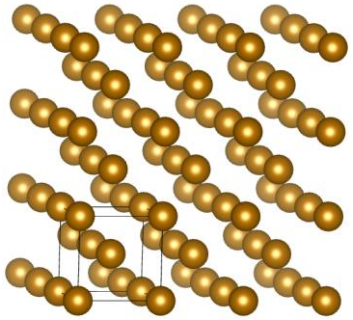
II. Space Group of Crystals

III. Space Partitioning: The Wigner-Seitz Cell

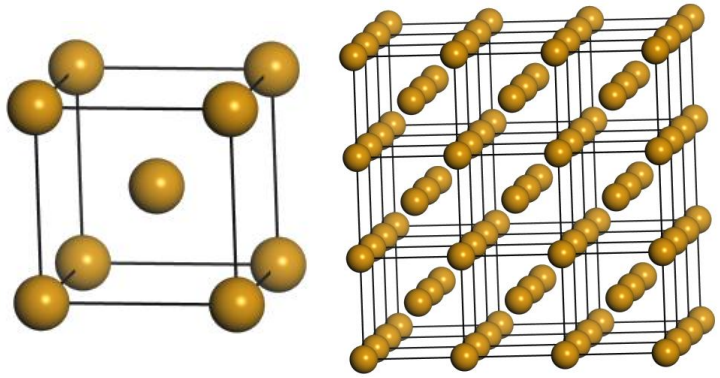
IV. Complex Real-World Crystals—Example: Graphene

I. The Geometry and Art of Real Space for Crystal Structures

Crystal Structure



Metal Iron Crystal



Unit Cell **Repeated Unit Cell**

The crystal structure consists of the same group of atoms, the basis, positioned around each and every lattice point. This group of atoms therefore repeats indefinitely in three dimensions according to the arrangement of one of the Bravais lattices.

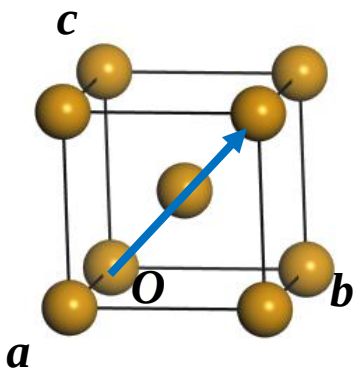
Bravais lattice, named after Auguste Bravais (1850), is an infinite array of discrete points generated by a set of discrete translation operations described in three dimensional space by

$$\boldsymbol{R = n_1 a_1 + n_2 a_2 + n_3 a_3}$$

where the n_i are any integers and a_i are primitive translation vectors, or primitive vectors, which lie in different directions (not necessarily mutually perpendicular) and span the lattice.

I. The Geometry and Art of Real Space for Crystal Structures

$$R = n_1 a_1 + n_2 a_2 + n_3 a_3$$

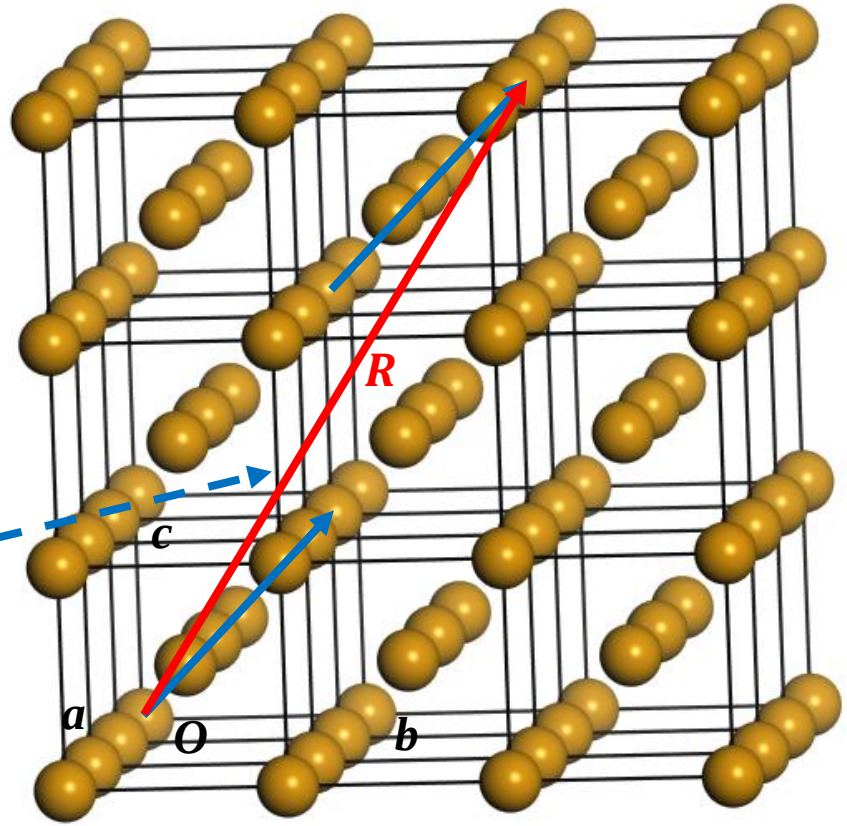


$n_1 = 2$

$n_2 = 2$

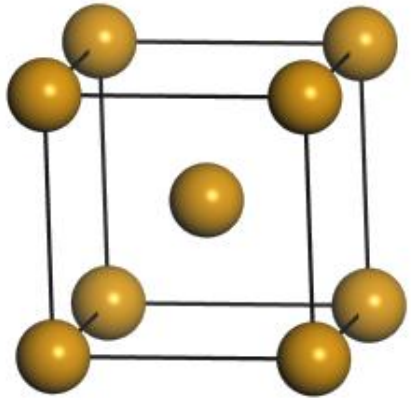
$n_3 = 3$

$R = 2a_1 + 2a_2 + 3a_3$

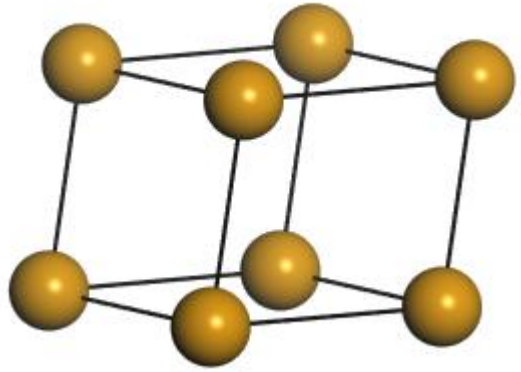


I. The Geometry and Art of Real Space for Crystal Structures

Unit Cell for Metal Iron Crystal



Conventional Cell



Primitive Cell

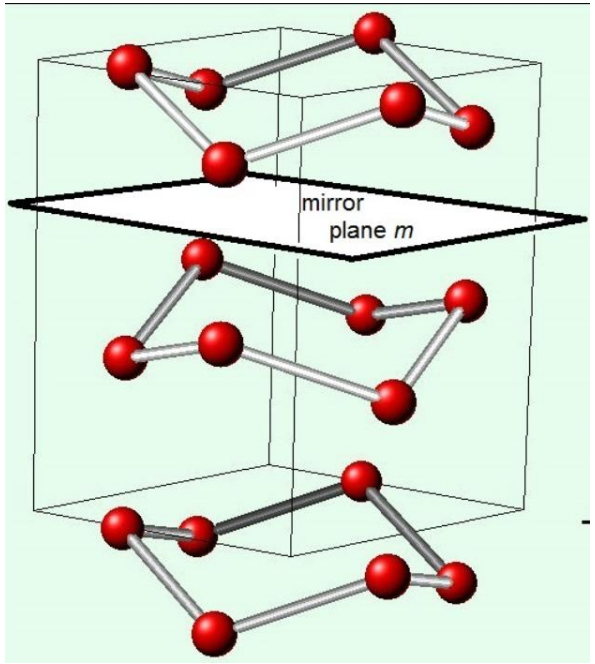
A larger cell chosen deliberately to highlight the geometric point-group symmetry of the crystal (*e.g.*, keeping things orthogonal at 90°)

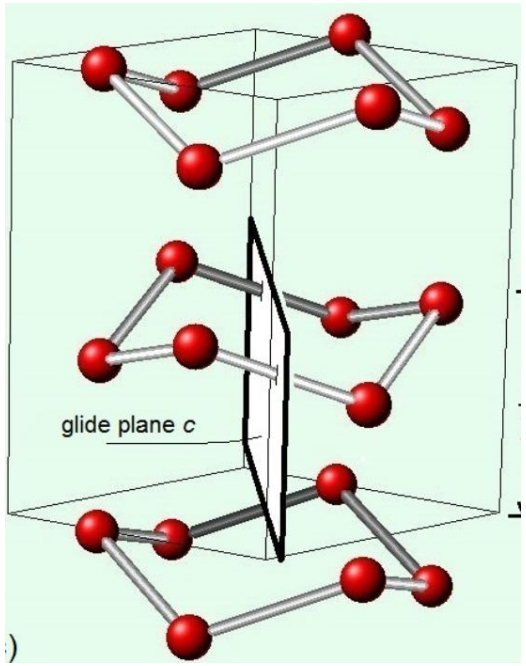
A primitive cell is a unit cell that contains exactly one lattice point. For unit cells generally, lattice points that are shared by n cells are counted as $^1/_n$ of the lattice points contained in each of those cells; so for example a primitive unit cell in three dimensions which has lattice points only at its eight vertices is considered to contain $^1/_8$of each of them

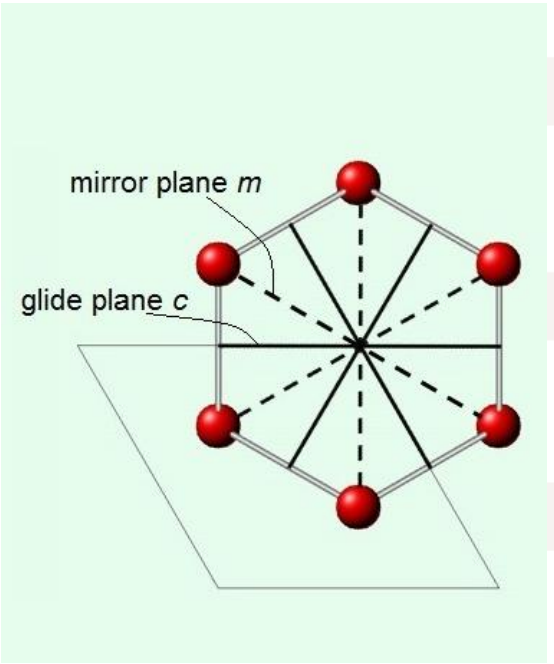
II. Space Group of Crystals

Space Group of Crystal

The symmetry group of a repeating pattern in space, usually in three dimensions. The elements of a space group (its symmetry operations) are the rigid transformations of the pattern that leave it unchanged.



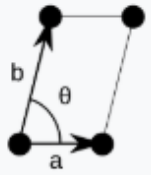
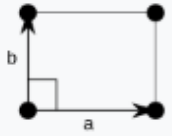
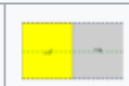
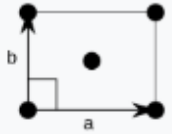
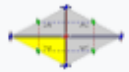
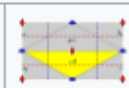
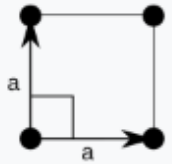
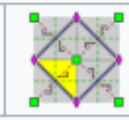
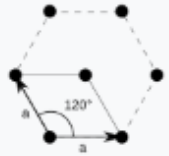
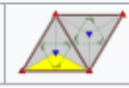
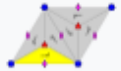
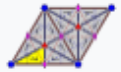




Special Symmetry → **System Unchanged** → **Properties remain same**

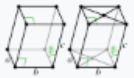
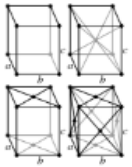
II. Space Group of Crystals

Space group to 2D system

Crystal system, Bravais lattice	Geometric class, point group					Arithmetic class	Wallpaper groups (cell diagram)				
	Int'l	Schön.	Orbifold	Cox.	Ord.						
Oblique 	1	C_1	(1)	$[\,]^+$	1	None	p1 (1)				
	2	C_2	(22)	$[2]^+$	2	None	p2 (2222)				
Rectangular 	m	D_1	(*)	$[\,]$	2	Along	pm (**)		pg (××)		
	2mm	D_2	(*22)	[2]	4	Along	pmm (*2222)		pmg (22*)		
Centered rectangular 	m	D_1	(*)	$[\,]$	2	Between	cm (*×)				
	2mm	D_2	(*22)	[2]	4	Between	cmm (2*22)		pgg (22×)		
Square 	4	C_4	(44)	$[4]^+$	4	None	p4 (442)				
	4mm	D_4	(*44)	[4]	8	Both	p4m (*442)		p4g (4*2)		
Hexagonal 	3	C_3	(33)	$[3]^+$	3	None	p3 (333)				
	3m	D_3	(*33)	[3]	6	Between	p3m1 (*333)		p31m (3*3)		
	6	C_6	(66)	$[6]^+$	6	None	p6 (632)				
	6mm	D_6	(*66)	[6]	12	Both	p6m (*632)				

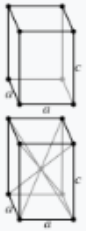
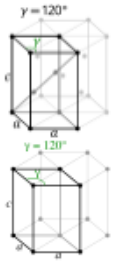
II. Space Group of Crystals

Space group to 3D system

№	Crystal system, (count), Bravais lattice	Point group					Space groups (international short symbol)
		Int'l	Schön.	Orbifold	Cox.	Ord.	
1	Triclinic (2) 	1	C_1	11	$[\,]^+$	1	P1
2		$\bar{1}$	C_i	1×	$[2^+,2^+]$	2	$P\bar{1}$
3–5	Monoclinic (13) 	2	C_2	22	$[2]^+$	2	$P2$, $P2_1$ $C2$
6–9		m	C_s	*11	$[\,]$	2	Pm, Pc Cm, Cc
10–15		2/m	C_{2h}	2*	$[2,2^+]$	4	$P2/m$, $P2_1/m$ $C2/m$, $P2/c$, $P2_1/c$ $C2/c$
16–24	Orthorhombic (59) 	222	D_2	222	$[2,2]^+$	4	$P222$, $P222_1$, $P2_12_12$, $P2_12_12_1$ $C222_1$, $C222$ $F222$ $I222$, $I2_12_12_1$
25–46		mm2	C_{2v}	*22	[2]	4	$Pmm2$, $Pmc2_1$, $Pcc2$, $Pma2$, $Pca2_1$, $Pnc2$, $Pmn2_1$, $Pba2$, $Pna2_1$, $Pnn2$ $Cmm2$, $Cmc2_1$, $Ccc2$, $Amm2$, $Aem2$, $Ama2$, $Aea2$ $Fmm2$, $Fdd2$ $Imm2$, $Iba2$, $Ima2$
47–74		mmm	D_{2h}	*222	[2,2]	8	$Pmmm$, $Pnnn$, $Pccm$, $Pban$, $Pmma$, $Pnna$, $Pmna$, $Pcca$, $Pbam$, $Pccn$, $Pbcm$, $Pnnm$, $Pmmn$, $Pbcn$, $Pbca$, $Pnma$ $Cmcm$, $Cmce$, $Cmmm$, $Cccm$, $Cmme$, $Ccce$ $Fmmm$, $Fddd$ $Immm$, $Ibam$, $Ibca$, $Imma$

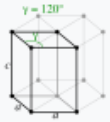
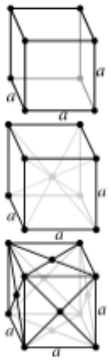
II. Space Group of Crystals

Space group to 3D system

75–80	Tetragonal (68) 	4	C_4	44	$[4]^+$	4	P4, $P4_1$, $P4_2$, $P4_3$, I4, $I4_1$
81–82		$\bar{4}$	S_4	2×	$[2^+,4^+]$	4	P$\bar{4}$, I$\bar{4}$
83–88		4/m	C_{4h}	4*	$[2,4^+]$	8	P4/m, $P4_2$/m, P4/n, $P4_2$/n I4/m, $I4_1$/a
89–98		422	D_4	224	$[2,4]^+$	8	P422, $P42_12$, $P4_122$, $P4_12_12$, $P4_222$, $P4_22_12$, $P4_322$, $P4_32_12$ I422, $I4_122$
99–110		4mm	C_{4v}	*44	[4]	8	P4mm, P4bm, $P4_2$cm, $P4_2$nm, P4cc, P4nc, $P4_2$mc, $P4_2$bc I4mm, I4cm, $I4_1$md, $I4_1$cd
111–122		$\bar{4}$2m	D_{2d}	2*2	$[2^+,4]$	8	P$\bar{4}$2m, P$\bar{4}$2c, P$\bar{4}2_1$m, P$\bar{4}2_1$c, P$\bar{4}$m2, P$\bar{4}$c2, P$\bar{4}$b2, P$\bar{4}$n2 I$\bar{4}$m2, I$\bar{4}$c2, I$\bar{4}$2m, I$\bar{4}$2d
123–142		4/mmm	D_{4h}	*224	[2,4]	16	P4/mmm, P4/mcc, P4/nbm, P4/nnc, P4/mbm, P4/mnc, P4/nmm, P4/ncc, $P4_2$/mmc, $P4_2$/mcm, $P4_2$/nbc, $P4_2$/nnm, $P4_2$/mbc, $P4_2$/mnm, $P4_2$/nmc, $P4_2$/ncm I4/mmm, I4/mcm, $I4_1$/amd, $I4_1$/acd
143–146	Trigonal (25) 	3	C_3	33	$[3]^+$	3	P3, $P3_1$, $P3_2$ R3
147–148		$\bar{3}$	S_6	3×	$[2^+,6^+]$	6	P$\bar{3}$, R$\bar{3}$
149–155		32	D_3	223	$[2,3]^+$	6	P312, P321, $P3_112$, $P3_121$, $P3_212$, $P3_221$ R32
156–161		3m	C_{3v}	*33	[3]	6	P3m1, P31m, P3c1, P31c R3m, R3c
162–167		$\bar{3}$m	D_{3d}	2*3	$[2^+,6]$	12	P$\bar{3}$1m, P$\bar{3}$1c, P$\bar{3}$m1, P$\bar{3}$c1 R$\bar{3}$m, R$\bar{3}$c

II. Space Group of Crystals

Space group to 3D system

168–173	Hexagonal (27) 	6	C_6	66	$[6]^+$	6	P6, $P6_1$, $P6_5$, $P6_2$, $P6_4$, $P6_3$
174		$\bar{6}$	C_{3h}	3*	$[2,3^+]$	6	$P\bar{6}$
175–176		6/m	C_{6h}	6*	$[2,6^+]$	12	P6/m, $P6_3/m$
177–182		622	D_6	226	$[2,6]^+$	12	P622, $P6_122$, $P6_522$, $P6_222$, $P6_422$, $P6_322$
183–186		6mm	C_{6v}	*66	[6]	12	P6mm, P6cc, $P6_3cm$, $P6_3mc$
187–190		$\bar{6}$m2	D_{3h}	*223	[2,3]	12	P$\bar{6}$m2, P$\bar{6}$c2, P$\bar{6}$2m, P$\bar{6}$2c
191–194		6/mmm	D_{6h}	*226	[2,6]	24	P6/mmm, P6/mcc, $P6_3/mcm$, $P6_3/mmc$
195–199	Cubic (36) 	23	T	332	$[3,3]^+$	12	P23, F23, I23 $P2_13$, $I2_13$
200–206		m$\bar{3}$	T_h	3*2	$[3^+,4]$	24	Pm$\bar{3}$, Pn$\bar{3}$, Fm$\bar{3}$, Fd$\bar{3}$, Im$\bar{3}$, Pa$\bar{3}$, Ia$\bar{3}$
207–214		432	O	432	$[3,4]^+$	24	P432, $P4_232$ F432, $F4_132$ I432 $P4_332$, $P4_132$, $I4_132$
215–220		$\bar{4}$3m	T_d	*332	[3,3]	24	P$\bar{4}$3m, F$\bar{4}$3m, I$\bar{4}$3m P$\bar{4}$3n, F$\bar{4}$3c, I$\bar{4}$3d
221–230		m$\bar{3}$m	O_h	*432	[3,4]	48	Pm$\bar{3}$m, Pn$\bar{3}$n, Pm$\bar{3}$n, Pn$\bar{3}$m Fm$\bar{3}$m, Fm$\bar{3}$c, Fd$\bar{3}$m, Fd$\bar{3}$c Im$\bar{3}$m, Ia$\bar{3}$d

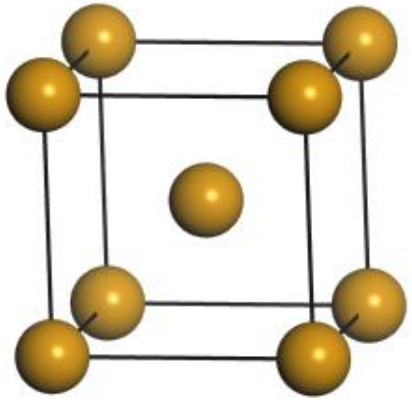
II. Space Group of Crystals

Wycoff Position

In crystallography, a Wyckoff position is any point in a set of points whose site symmetry groups (see below) are all conjugate subgroups of one another.

Crystallography tables give the Wyckoff positions for different space groups.

Atoms in a unit cell can be described by just their wyckoff position and symmetry operators of the space group can be used to determine the other coordinates within a unit cell.



Unit Cell for Metal Iron Crystal

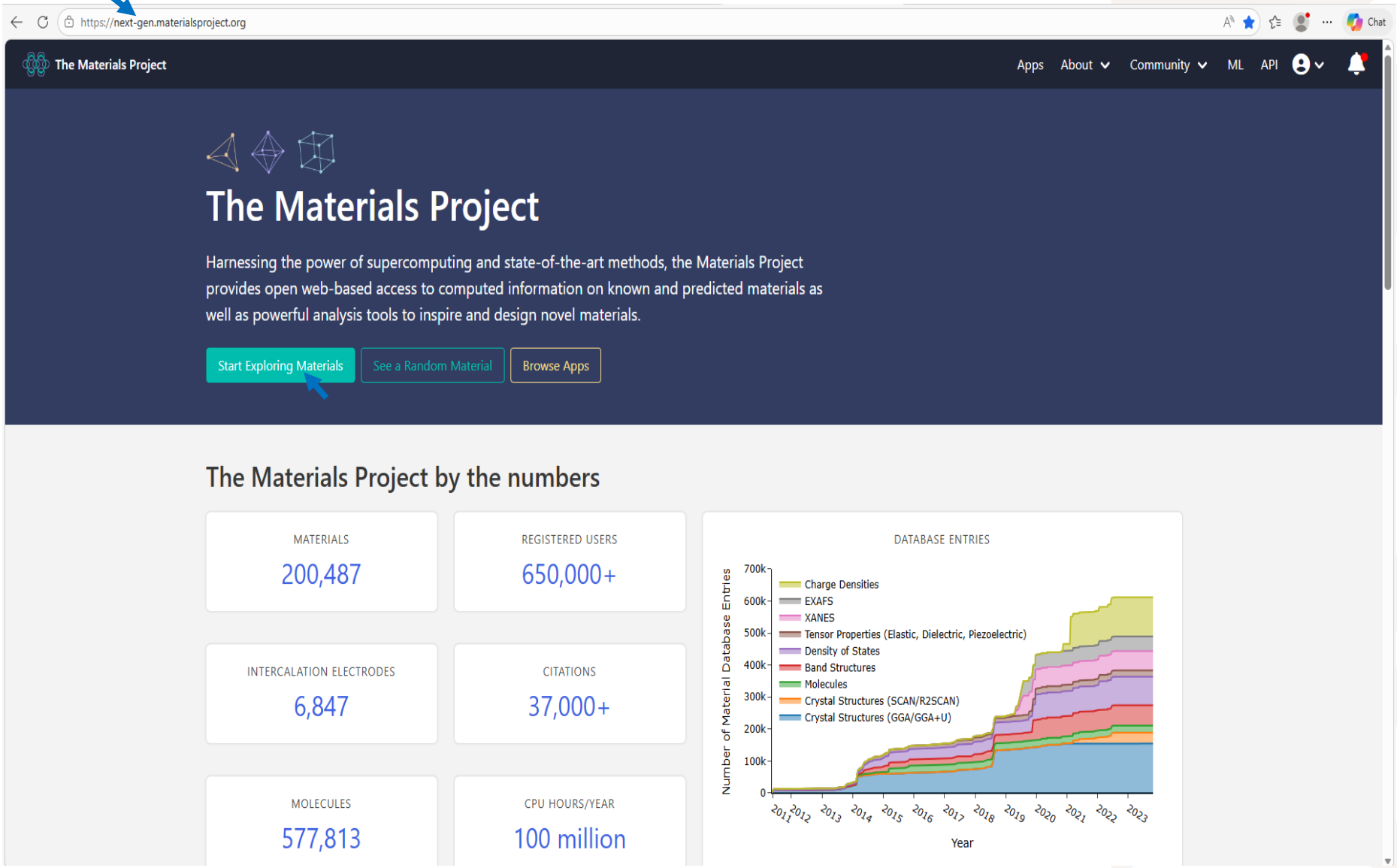
Wyckoff Positions

Label	Element	a	b	c
2a	Fe	0	0	0

II. Space Group of Crystals

https://next-gen.materialsproject.org/ **How to build up crystal structure**

Step One

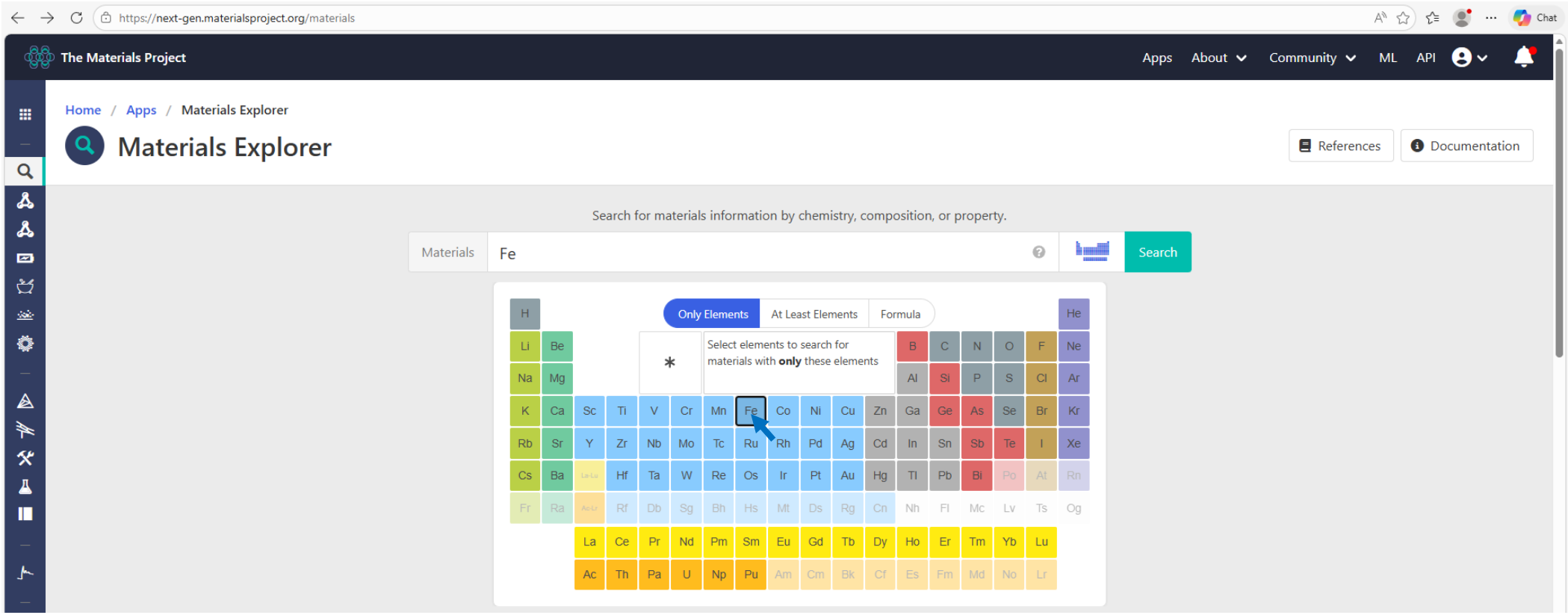


II. Space Group of Crystals

How to build up crystal structure

Step Two

Choose all elements the crystal has

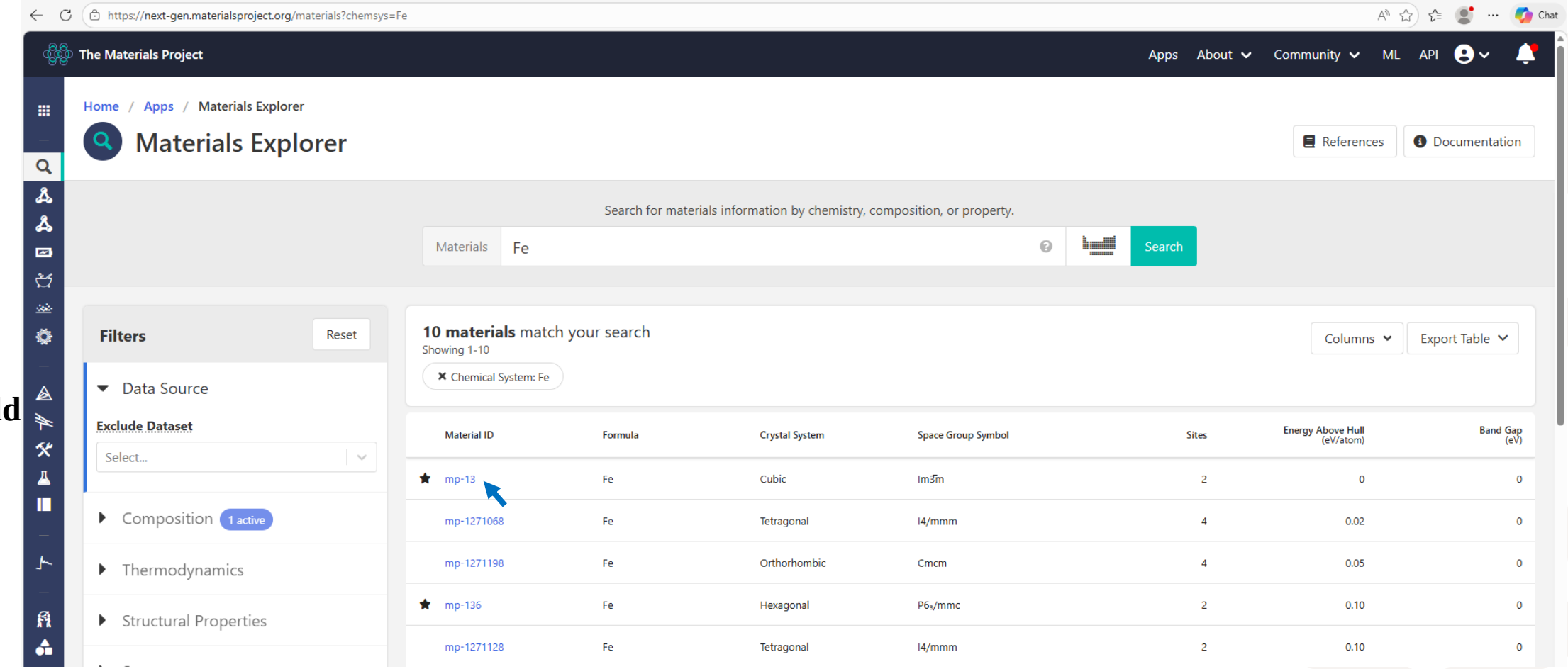


II. Space Group of Crystals

How to build up crystal structure

Step Three

Choose the crystal to build

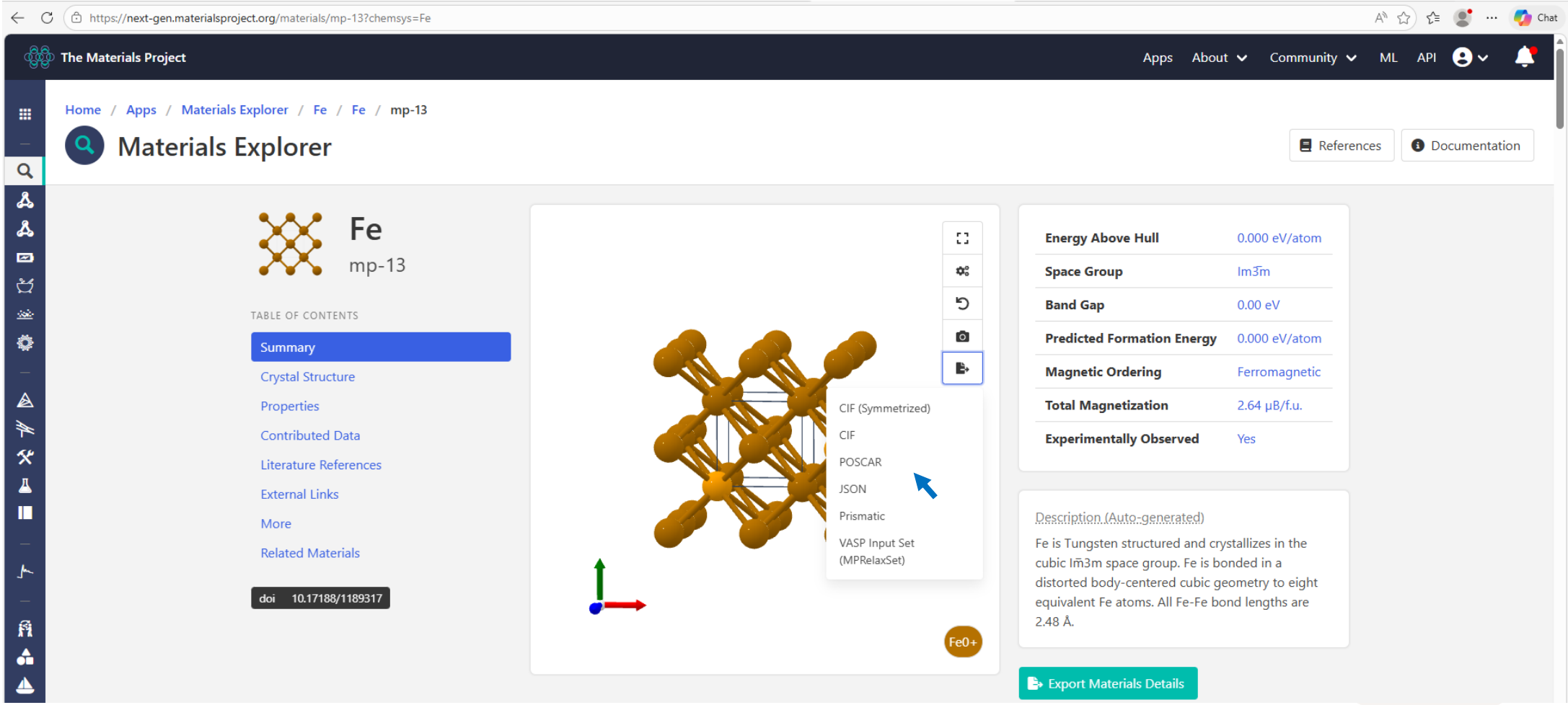


II. Space Group of Crystals

How to build up crystal structure

Step Four

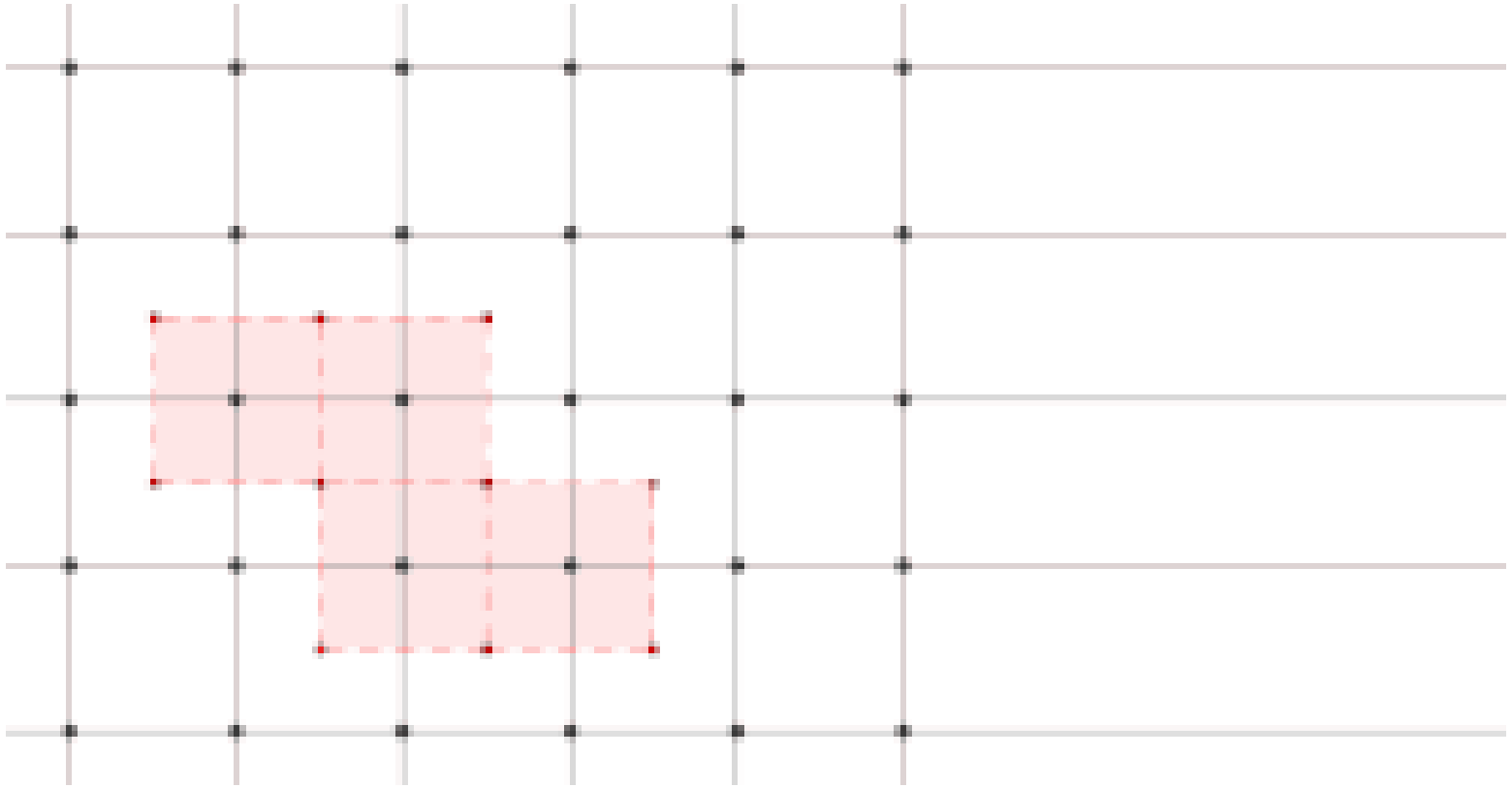
Export structure



III. Space Partitioning: The Wigner-Seitz Cell

A Wigner–Seitz cell is an example of a primitive cell, which is a unit cell containing exactly one lattice point

It is the locus of points in space that are closer to that lattice point than to any of the other lattice points



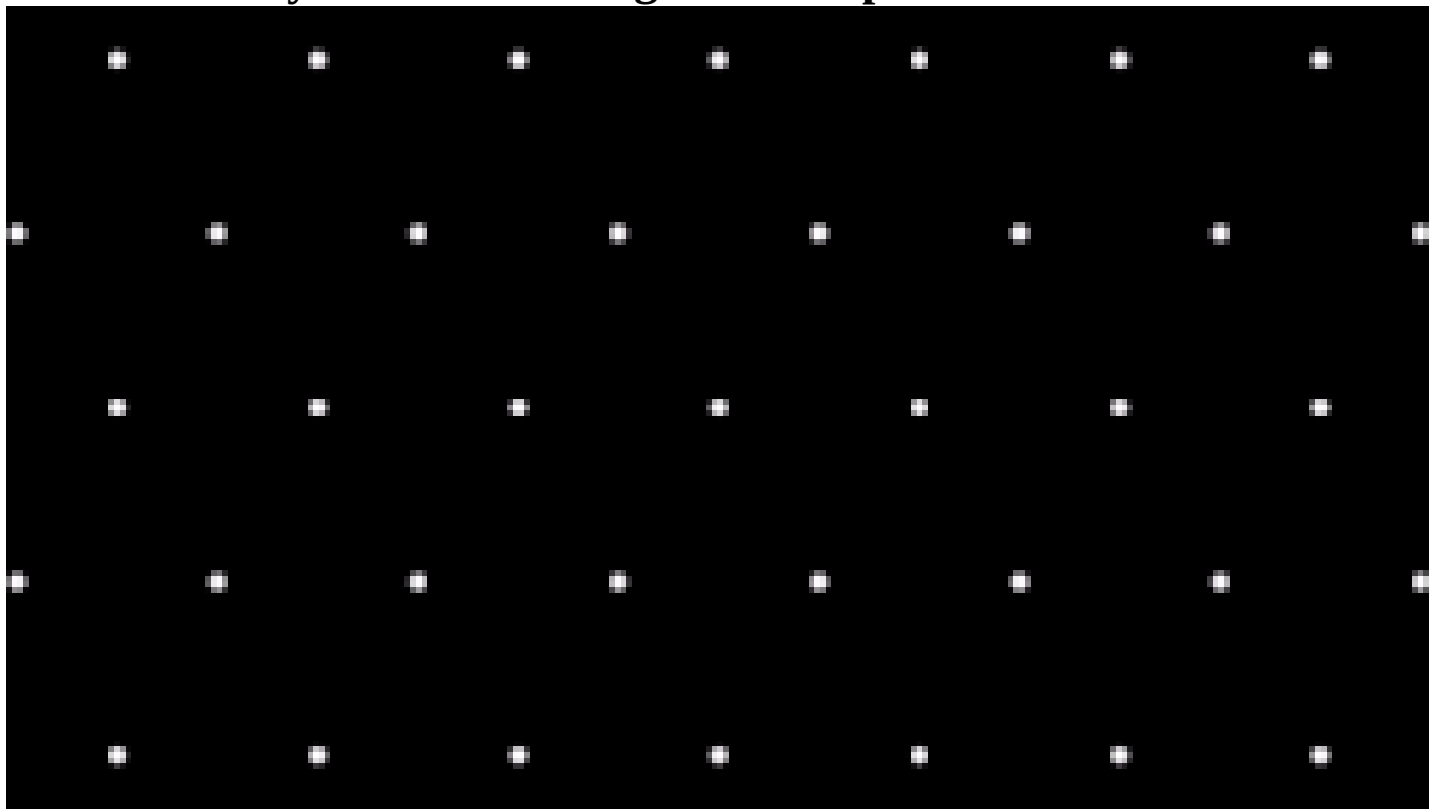
III. Space Partitioning: The Wigner-Seitz Cell

The cell can be chosen by first picking a lattice point

After a point is chosen, lines are drawn to all nearby lattice points

At the midpoint of each line, another line is drawn normal to each of the first set of lines

The smallest area enclosed in this way is called the Wigner–Seitz primitive cell

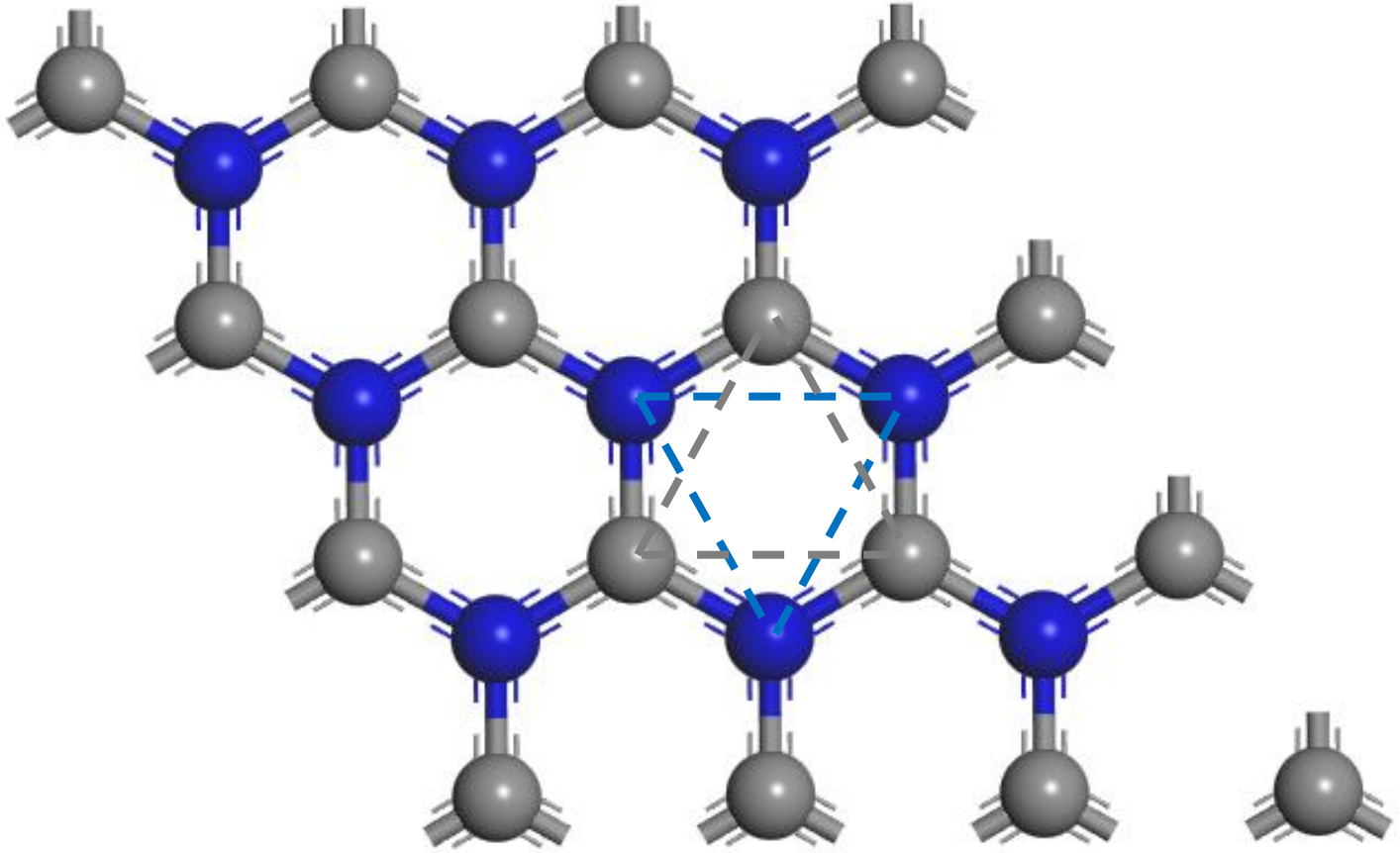


IV. Complex Real-World Crystals—Example: Graphene

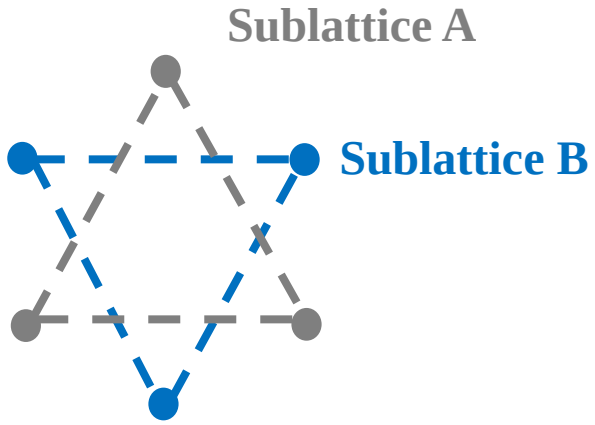
Non-Bravais Lattices

A compound lattice (more commonly referred to in English as a Lattice with a Basis or a Non-Bravais Lattice) is a crystal structure where the repeating unit consists of more than one atom

Pristine Graphene



IV. Complex Real-World Crystals—Example: Graphene



Pristine Graphene

♦ **An *A*-type atom has its three nearest-neighbor chemical bonds pointing Up, Down-Left, and Down-Right**

♦ **A *B*-type atom has its bonds pointing Down, Up-Left, and Up-Right**

Because their local spatial environments are rotated by 180°, one cannot reach a *B*-site from an *A*-site by a pure lattice translation vector *R*

Therefore, they belong to a compound lattice consisting of a simple 2D hexagonal Bravais lattice populated by a two-carbon basis



That's all for this session

Thank you for your attention and great participation